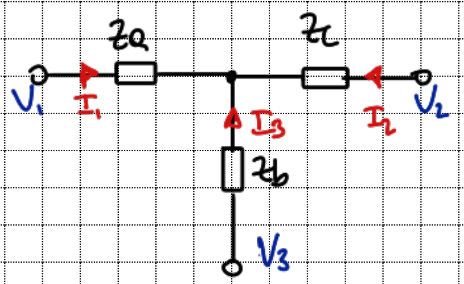
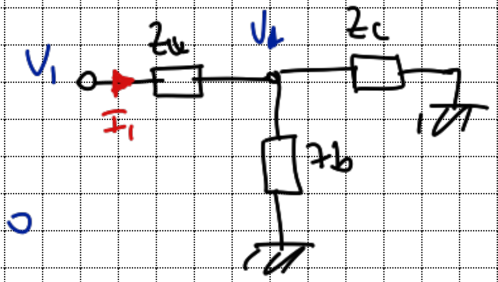
# Tarea 6 - Maximiliano Montenegro

1-




$$Y_{11} = \left.\frac{I_1}{V_1}\right|_{V_2 = V_3 = 0}$$




$$V_A(Y_a+Y_b+Y_c) = V_1 Y_a$$

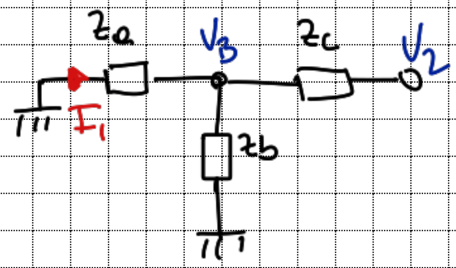
$$V_A = V_1 \frac{Y_a}{Y_a+Y_b+Y_c}$$

$$I_1 = V_1 - V_A$$

$$I_1 = V_1\left(1 - \frac{Y_a}{Y_a+Y_b+Y_c}\right) Y_a$$

$$\frac{I_1}{V_1} = \frac{(Y_b+Y_c)Y_a}{Y_a+Y_b+Y_c} = \boxed{Y_{11} = \frac{Y_aY_b + Y_aY_c}{Y_a+Y_b+Y_c}}$$

$$Y_{12} = \left.\frac{I_1}{V_2}\right|_{V_1 = V_3 = 0}$$




$$V_B(Y_a+Y_b+Y_c) = V_2 Y_c$$

$$V_B = V_2 \frac{Y_c}{Y_a+Y_b+Y_c}$$

$$I_1 = -V_B Y_a$$

$$\frac{I_1}{V_2} = \boxed{\frac{-Y_aY_c}{Y_a+Y_b+Y_c} = Y_{12}}$$

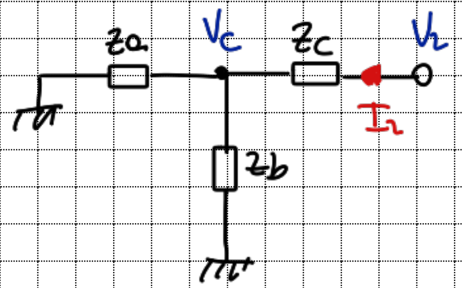
Por propiedad de la MAI $\sum filas = 0$

$$\Rightarrow Y_{11} + Y_{12} + Y_{13} = 0$$

$$Y_aY_b + \cancel{Y_aY_c} - \cancel{Y_aY_c} + Y_{13} = 0 \rightarrow Y_{13} = -Y_aY_b$$

$$\boxed{Y_{13} = -\frac{Y_aY_b}{Y_a+Y_b+Y_c}} \qquad Y_{12} = Y_{21} \quad \wedge \quad Y_{13} = Y_{31}$$

$$Y_{22} = \left.\frac{V_2}{I_2}\right| \quad V_1 = V_3 = 0$$




$$V_C(Y_a+Y_b+Y_c) = V_2 Y_b$$

$$V_C = V_2 \frac{Y_C}{Y_a+Y_b+Y_C}$$

$$I_2 = (V_2 - V_C) Y_C$$

$$I_2 = V_2 \left(1 - \frac{Y_c}{Y_a+Y_b+Y_C}\right) Y_c$$

$$\frac{I_2}{V_2} = \frac{(Y_a + Y_b + \cancel{Y_C} - \cancel{Y_C}) Y_c}{Y_a+Y_b+Y_C}$$

$$Y_{22} = \frac{Y_aY_c + Y_bY_c}{Y_a+Y_b+Y_c}$$

Utilizo la propiedad de columna y fila

$$Y_{21} + Y_{22} + Y_{23} = 0$$

$$-\cancel{Y_aY_c} + \cancel{Y_aY_C} + Y_bY_C + Y_{23} = 0$$

$$\Rightarrow Y_{23} = Y_{32} = \frac{-Y_bY_C}{Y_a+Y_b+Y_C}$$

$$\wedge \quad Y_{31} + Y_{32} + Y_{33} = 0$$

$$-Y_aY_b + -Y_bY_C + Y_{33} = 0$$

$$Y_{33} = \frac{Y_aY_b + Y_bY_C}{Y_a+Y_b+Y_C}$$

$$MAI: \frac{1}{Y_a+Y_b+Y_C}\begin{pmatrix} Y_a(Y_b+Y_c) & -Y_aY_c & -Y_aY_b \\ -Y_aY_C & Y_c(Y_a+Y_b) & -Y_bY_C \\ -Y_aY_b & -Y_bY_C & Y_b(Y_a+Y_c) \end{pmatrix}$$

$$Y_a = \frac{1}{SL} \; ; \; Y_B = SC \; ; \; Y_C = \frac{1}{SL}$$

$$Y_a+Y_b+Y_c = \frac{1}{SL} + SC + \frac{1}{SL} = 2\frac{1}{SL} + SC = \frac{2+S^2LC}{SL} = Y_a+Y_b+Y_c$$

$$\frac{1}{Y_a+Y_b+Y_c} = \frac{SL}{S^2LC+2}$$

$$\bullet\ Y_a(Y_b+Y_c) = \frac{1}{SL}\left(SC + \frac{1}{SL}\right) = \frac{1}{SL}\left(\frac{S^2LC+1}{SL}\right)$$

$$\frac{Y_a(Y_b+Y_c)}{Y_a+Y_b+Y_c} = \frac{\cancel{SL}}{S^2LC+2} \cdot \frac{S^2LC+1}{(SL)^{\cancel{2}}} = \frac{S^2LC+1}{SL(S^2LC+2)} = Y_{11}$$

$$\bullet\ -Y_a . Y_c = -\frac{1}{SL} \cdot \frac{1}{SL} = -\left(\frac{1}{SL}\right)^2$$

$$\frac{-Y_aY_c}{Y_a+Y_b+Y_c} = \left(-\frac{1}{SL}\right)^2 \cdot \frac{SL}{S^2LC+2} = -\frac{1}{SL(S^2LC+2)} = Y_{12} = Y_{21}$$

$$\bullet\ -Y_a Y_b = -\frac{1}{SL} . SC = -\frac{C}{L}$$

$$\frac{-Y_bY_c}{Y_a+Y_b+Y_c} = \frac{SL}{S^2LC+2} \cdot -\frac{C}{L} = -\frac{SC}{S^2LC+2} = Y_{13} = Y_{31}$$

$$\bullet\ Y_c(Y_a+Y_b) = \frac{1}{SL}\left(\frac{1}{SL} + SC\right) = Y_a(Y_b+Y_c)$$

$$\Rightarrow \frac{Y_c(Y_a+Y_b)}{Y_a+Y_b+Y_c} = \frac{S^2LC+1}{SL(S^2LC+2)} = Y_{22}$$

$$\bullet\ -Y_bY_c = -SC . \frac{1}{SL} = -\frac{C}{L}$$

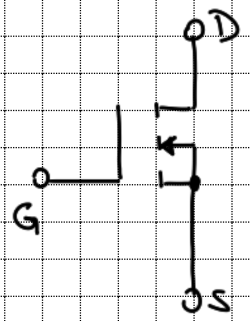
$$\frac{-Y_bY_c}{Y_a+Y_b+Y_c} = \frac{-SC}{S^2LC+2} = Y_{23} = Y_{32}$$

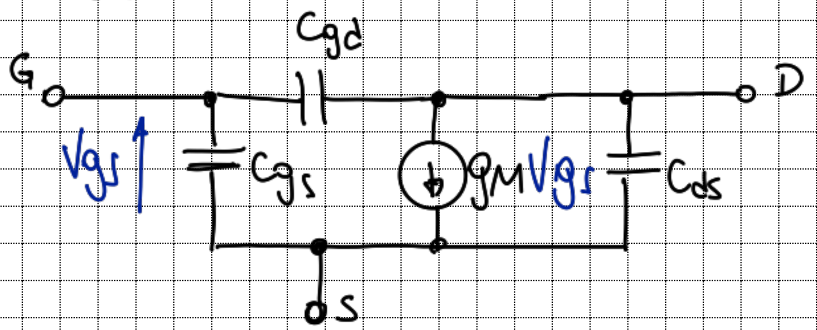
$$\bullet\ Y_b(Y_a+Y_c) = SC\left(\frac{1}{SL} + \frac{1}{SL}\right) = SC\left(\frac{2}{SL}\right) = 2\frac{C}{L}$$

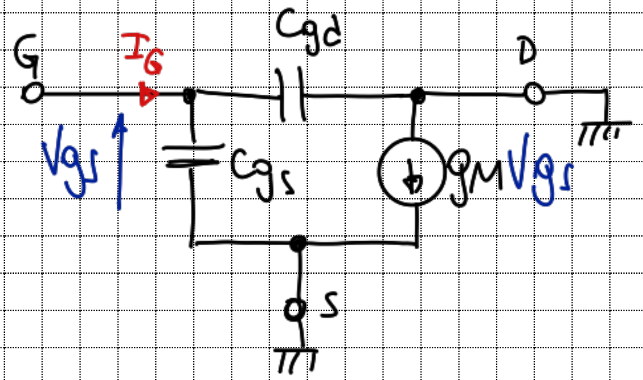
$$\frac{Y_b(Y_a+Y_c)}{Y_a+Y_b+Y_c} = \frac{SL}{S^2LC+2} \cdot 2\frac{C}{L} \Rightarrow \frac{2SC}{S^2LC+2} = Y_{33}$$

$$MAI = \begin{pmatrix} \frac{S^2LC+1}{SL(S^2LC+2)} & \frac{-1}{SL(S^2LC+2)} & -\frac{SC}{S^2LC+2} \\ \frac{-1}{SL(S^2LC+2)} & \frac{S^2LC+1}{SL(S^2LC+2)} & \frac{-SC}{S^2LC+2} \\ -\frac{SC}{S^2LC+2} & \frac{-SC}{S^2LC+2} & \frac{2SC}{S^2LC+2} \end{pmatrix}$$

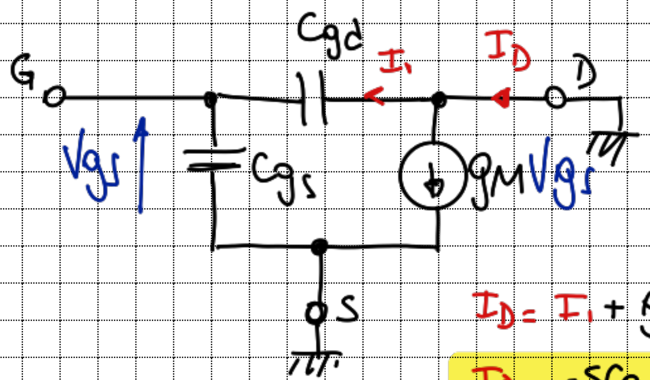
( BS170)



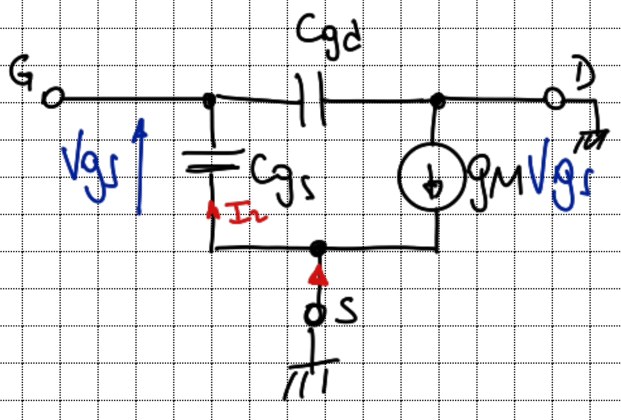

$\frac{I_G}{V_g} = S(C_{gs} + C_{gd}) = Y_{11}$




$Y_{21} = \frac{I_D}{V_g}\Big|_{V_D = V_S = 0}$

$I_D = I_1 + g_m V_g$ $\quad I_1 = -SC_{gd}V_g$

$\frac{I_D}{V_g} = -SC_{gd} + g_m = Y_{21}$

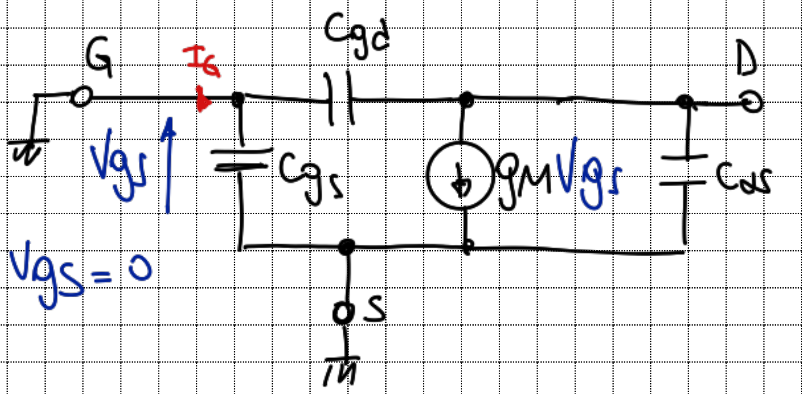



$Y_{31} = \frac{I_S}{V_g}$

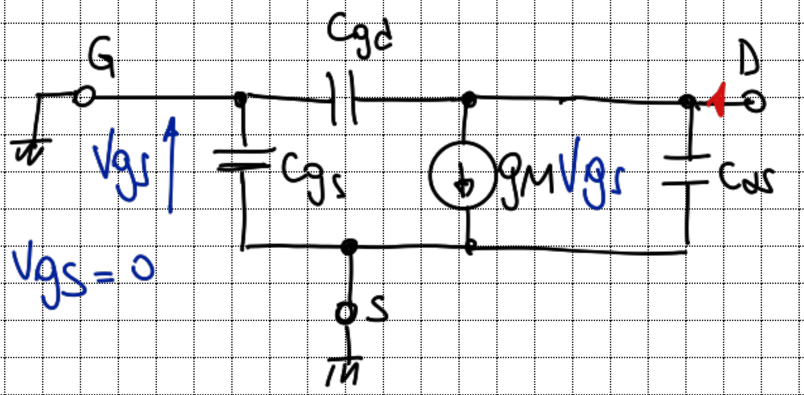
$I_2 = I_S + g_m V_g$

$I_2 = -SC_{gs}V_g$

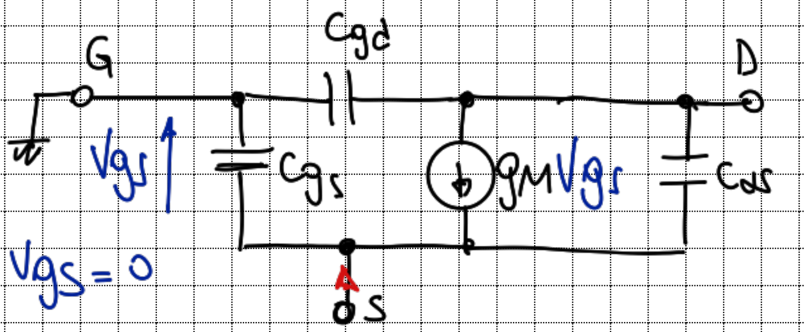
$\frac{I_S}{V_g} = -g_m - SC_{gs} = Y_{31}$



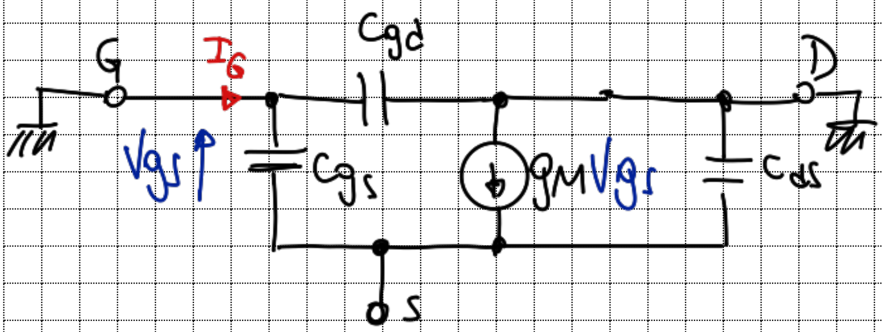

$$\frac{I_G}{V_D} = -sC_{gd} = Y_{12} =$$



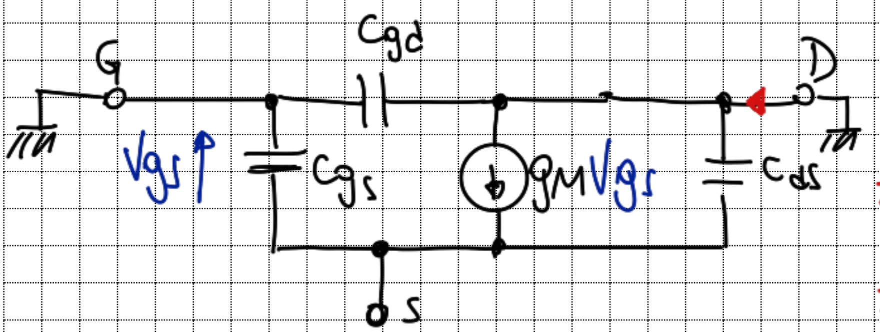

$$\frac{I_D}{V_D} = s(C_{gd} + C_{ds}) = Y_{22}$$




$$\frac{I_S}{V_D} = -sC_{ds} = Y_{32}$$



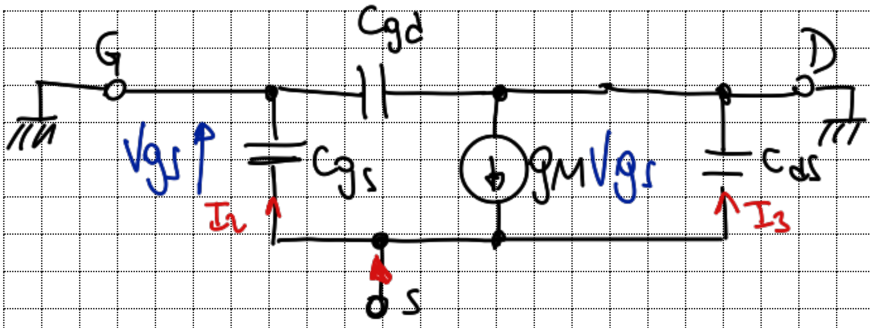

$$\frac{I_G}{V_S} = -sC_{gs} = Y_{13}$$




$$\frac{I_D}{V_S} =$$

$$I_D = g_m(-V_S) + sC_{ds}(-V_S)$$

$$\frac{I_D}{V_S} = -g_m - sC_{ds}$$




$$\frac{I_s}{V_s} \qquad V_{gs} = V_g - V_s = -V_s$$

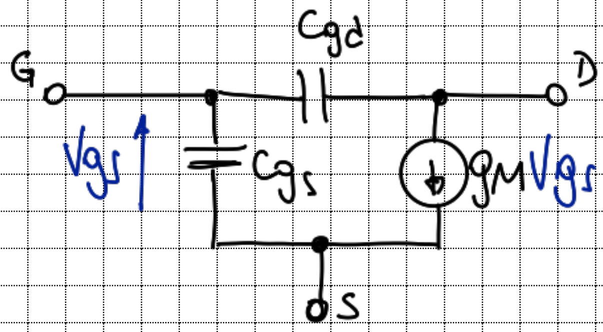
$$I_s = I_2 + I_3 - g_m(-V_s)$$

$$I_2 = sC_{gs}V_s$$

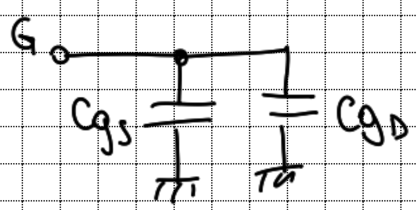
$$I_3 = sC_{ds}V_s$$

$$\frac{I_s}{V_s} = \boxed{s(C_{gs}+C_{ds}) + g_m}$$

$$MAI \begin{pmatrix} s(C_{gs}+C_{gd}) & -sC_{gd} & -sC_{gs} \\ g_m - sC_{gd} & s(C_{gd}+C_{ds}) & -g_m - sC_{ds} \\ -sC_{gs} - g_m & -sC_{ds} & s(C_{gs}+C_{ds})+g_m \end{pmatrix}$$

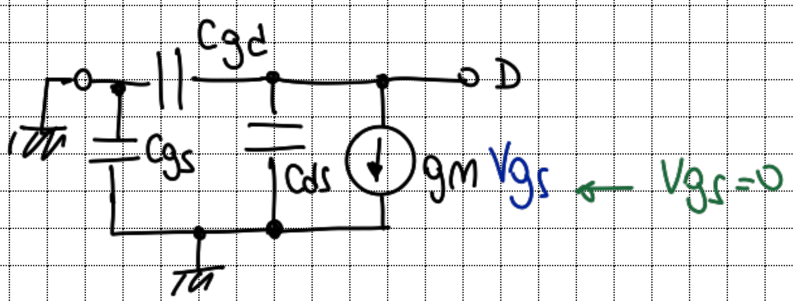



para $C_{iss}$
coloco D y S
a masa



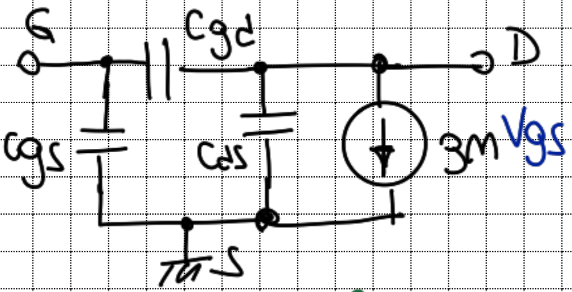

$$\boxed{C_{iss} = C_{gs} + C_{gd}}$$

para $C_{oss}$
colocο G y S
a masa




$$\boxed{C_{oss} = C_{gd} + C_{ds}}$$

para Crss coloco a masa S, y veo desde




$\boxed{C_{rss} = C_{gd}}$

$C_{gs} = C_{iss} - C_{rss}$

$C_{ds} = C_{oss} - C_{rss}$

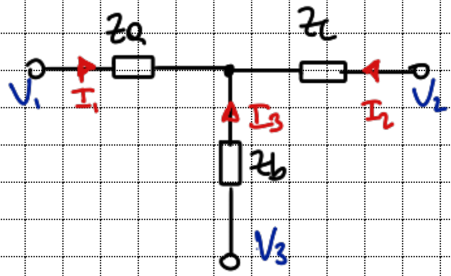
$$MAI \begin{pmatrix} s(C_{gs}+C_{gd}) & -sC_{gd} & -sC_{gs} \\ g_m - sC_{gd} & s(C_{gd}+C_{ds}) & -g_m - sC_{ds} \\ -sC_{gs} - g_m & -sC_{ds} & s(C_{gs}+C_{ds})+g_m \end{pmatrix}$$

$$\begin{pmatrix} sC_{iss} & -sC_{rss} & -s(C_{iss}-C_{rss}) \\ g_m - sC_{rss} & s(C_{oss}) & -g_m - s(C_{oss}-C_{rss}) \\ -g_m - s(C_{iss}-C_{rss}) & -s(C_{oss}-C_{rss}) & s(C_{iss}+C_{oss}-2C_{rss})+g_m \end{pmatrix}$$

2-a

$$MAI = \begin{pmatrix} \dfrac{s^2LC+1}{sL(s^2LC+2)} & \dfrac{-1}{sL(s^2LC+2)} & -\dfrac{sC}{s^2LC+2} \\ \dfrac{-1}{sL(s^2LC+2)} & \dfrac{s^2LC+1}{sL(s^2LC+2)} & \dfrac{-sC}{s^2LC+2} \\ -\dfrac{sC}{s^2LC+2} & \dfrac{-sC}{s^2LC+2} & \dfrac{2sC}{s^2LC+2} \end{pmatrix}$$

$$\left.\begin{aligned} \cdot\ V_{23} &= V_2 - V_3 \\ V_{13} &= V_1 - V_3 \end{aligned}\right\} \frac{V_{23}}{V_{13}} = \frac{V_2 - V_3}{V_1 - V_3}$$

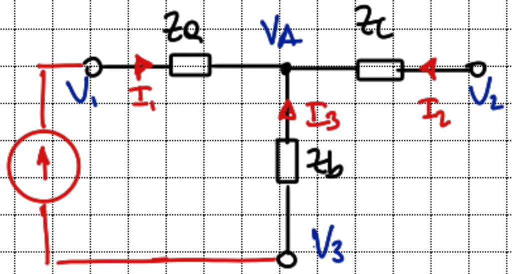



$$T^{ij}_{mn} = \frac{V_{ij}}{V_{mn}} = \frac{Y_{Mai\ ij}^{\ mn}}{Y_{Mai\ mn}^{\ mn}}$$

$$\Rightarrow \frac{V_{23}}{V_{13}} = \frac{Y_{Mai\ 23}^{\ 13}}{Y_{Mai\ 13}^{\ 13}} \cdot S \left\langle \begin{array}{l} (-1)^{i+j+m+n} \\ sg(i-j).sg(m-n) \end{array} \right.$$

$$\frac{V_{23}}{V_{13}} = \frac{\dfrac{-1}{sL(s^2LC+2)}}{\dfrac{s^2LC+1}{sL(s^2LC+2)}} = -\frac{1}{s^2LC+1} \cdot \left\langle \begin{array}{l} (-1)^{1+3+2+3} = -1 \\ sg(2-3)\,sg(1-3) = 1 \end{array} \right.$$

como difieren analizo el circuito




$$V_A - V_3 = (V_1 - V_3)\frac{Z_b}{Z_a + Z_b} \quad \Big|\ V_A = V_2$$

$$\frac{V_2 - V_3}{V_1 - V_3} = T^{23}_{13} = \frac{Z_b}{Z_a + Z_b}$$

$$T^{21}_{13} = \frac{1/sC}{sL + 1/sC} = \frac{1}{\dfrac{s^2LC+1}{sC}}\,\frac{1}{sC}$$

$Z_a = sL$

$Z_b = \dfrac{1}{sC}$

$Z_c = sL$

$$\boxed{T^{21}_{13} = \frac{1}{s^2LC+1}}$$

coincide con $(-1)^{1+3+2+3} = -1$

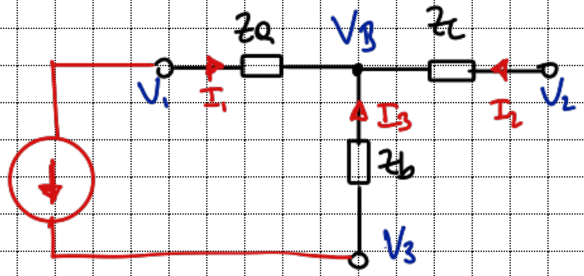
$$\frac{V_{21}}{V_{31}} = \frac{Y_{MAI\,21}^{\;31}}{Y_{MAI\,31}^{\;31}} = \frac{\dfrac{-sC}{s^2LC+2}}{\dfrac{s^2LC+1}{sL(s^2LC+2)}} = \frac{-sC}{sL(s^2LC+1)} \cdot$$

$$\frac{V_{21}}{V_{31}} = -\frac{s^2LC}{s^2LC+1}$$

$$(-1)^{3+1+2+1} = -1$$

$$sg(2-1)\,sg(3-1) = 1$$

como difieren en signo




$$V_B - V_1 = (V_3 - V_1)\frac{Z_a}{Z_a+Z_b} \quad \Big| \quad V_B = V_2$$

$$\frac{V_2 - V_1}{V_3 - V_1} = \frac{Z_a}{Z_a+Z_b}$$

$Z_a = sL$

$Z_b = \dfrac{1}{sC}$

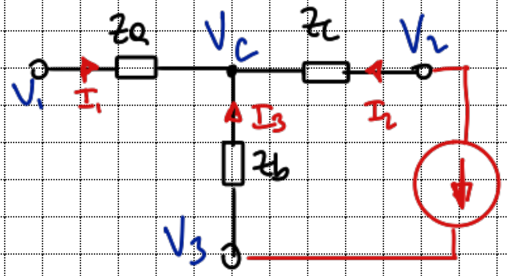
$Z_c = sL$

$$T_{31}^{21} = \frac{sL}{sL + \frac{1}{sC}} = \frac{s^2LC}{s^2LC+1} = T_{31}^{21}$$

coincide con $(-1)^{2+1+3+1} = -1$

$$\frac{V_{12}}{V_{32}} = \frac{Y_{MAI\,12}^{\;32}}{Y_{MAI\,32}^{\;32}} = \frac{\dfrac{-sC}{s^2LC+2}}{\dfrac{s^2LC+1}{sL(s^2LC+2)}} = -\frac{s^2LC}{s^2LC+1}\cdot S$$

$\underline{S}$: $(-1)^{1+2+3+2} = 1 \neq sg(1-2)\cdot sg(3-2) = -1$




$$V_C - V_2 = (V_3 - V_2)\frac{Z_c}{Z_b+Z_c} \quad \Big| \quad V_C = V_1$$

$$\frac{V_1 - V_2}{V_3 - V_2} = \frac{Z_c}{Z_b+Z_c}$$

$Z_a = sL$

$Z_b = \dfrac{1}{sC}$

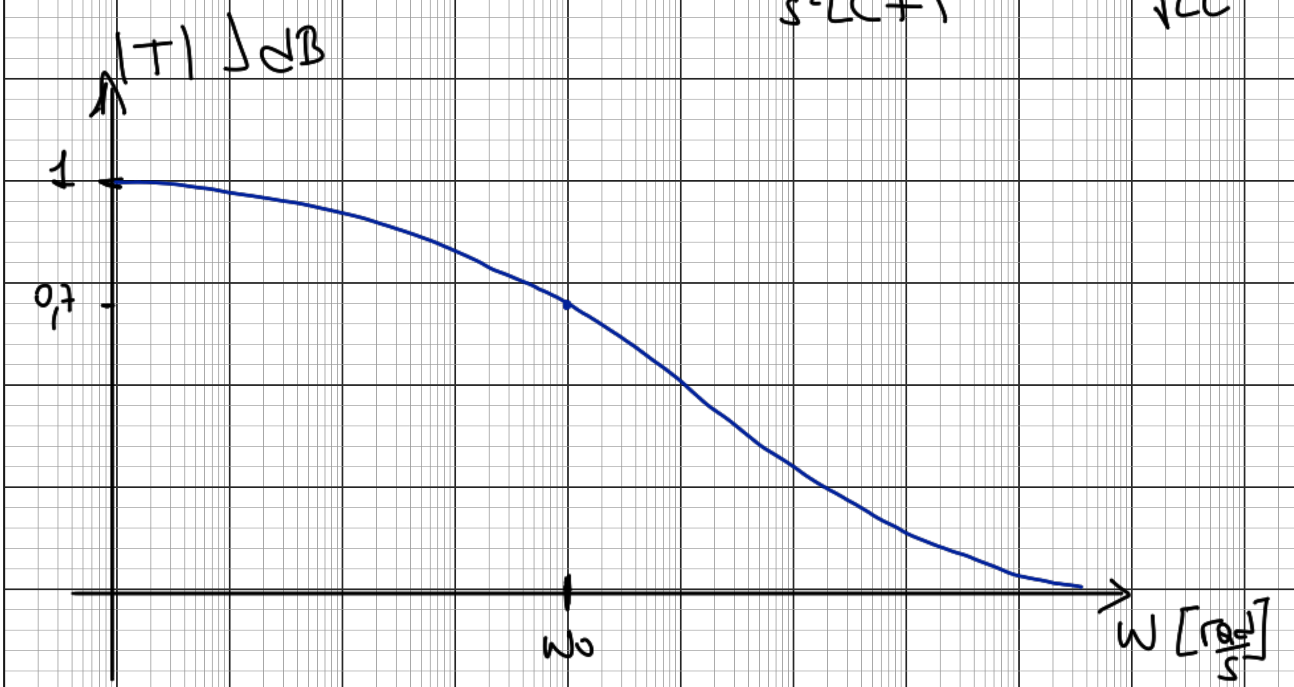
$Z_c = sL$

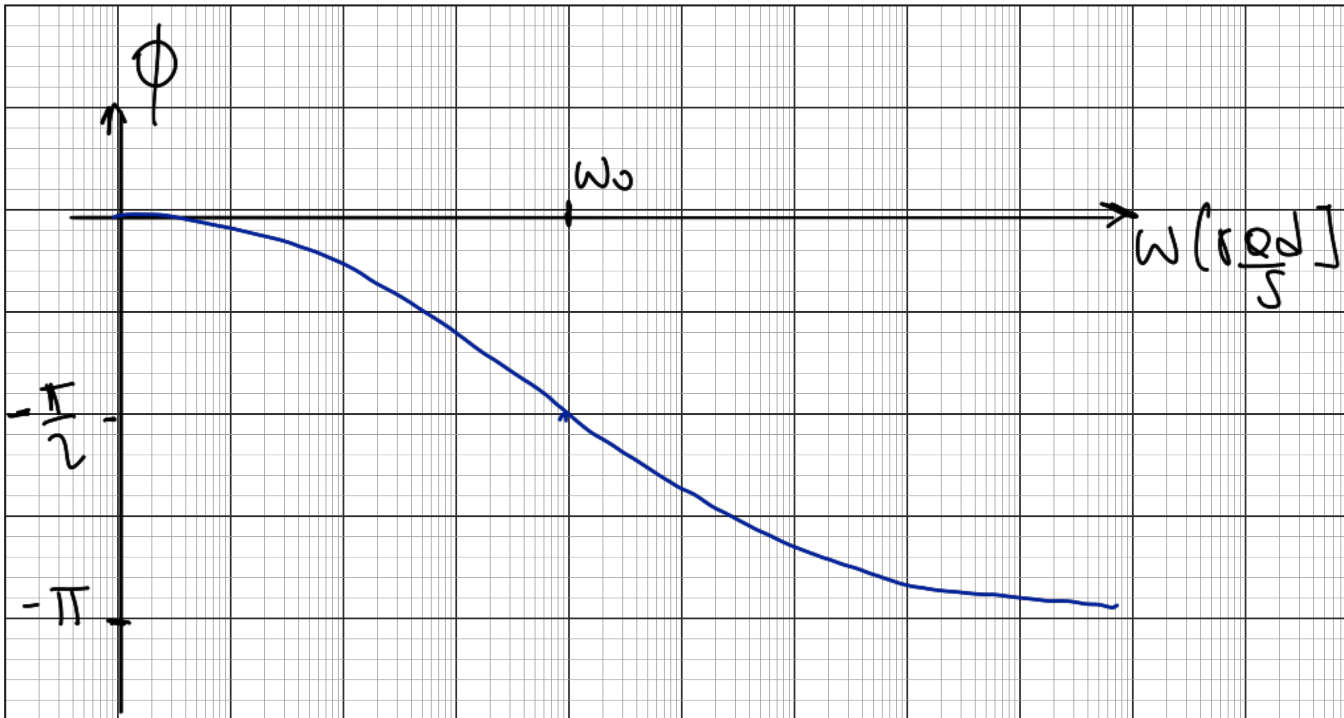
$$T_{32}^{12} = \frac{sL}{\frac{1}{sC} + sL} = \frac{s^2LC}{s^2LC+1} = T_{32}^{12}$$

coincide con $sg(1-2)\cdot sg(3-2) = -1$

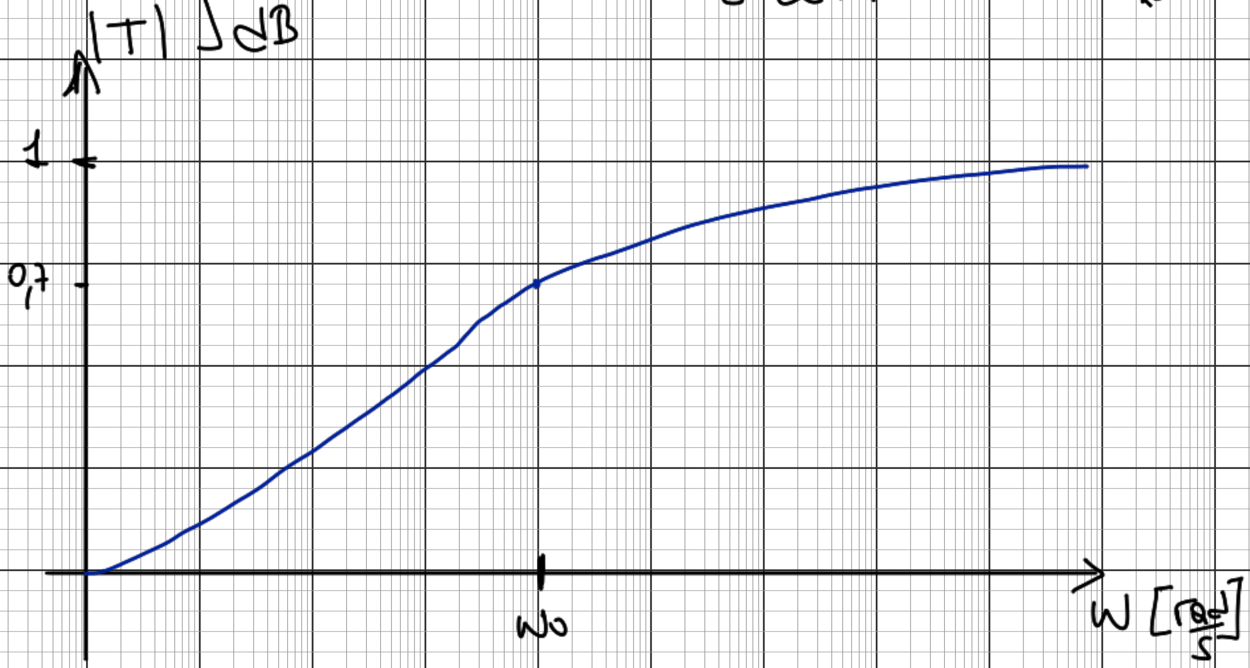
(b)

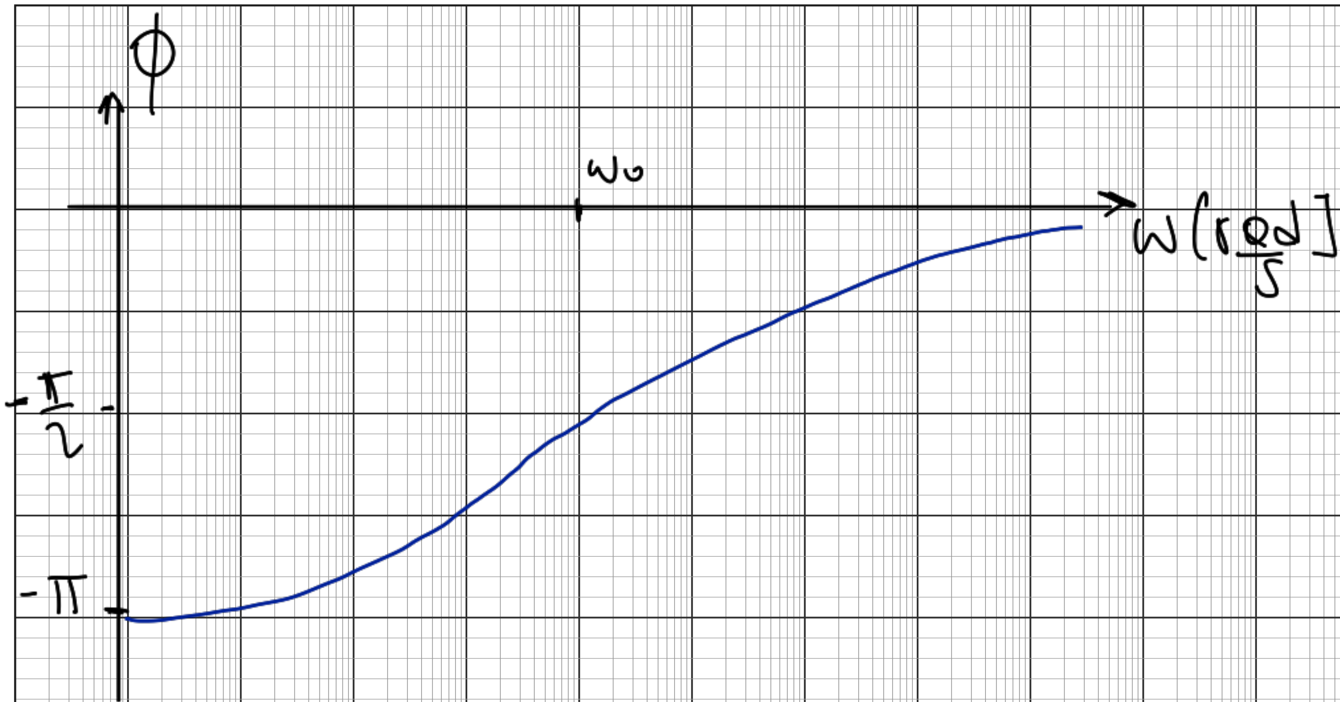
$$T_{13}^{21} = \frac{1}{s^2LC+1} \qquad \omega_0 = \frac{1}{\sqrt{LC}}$$



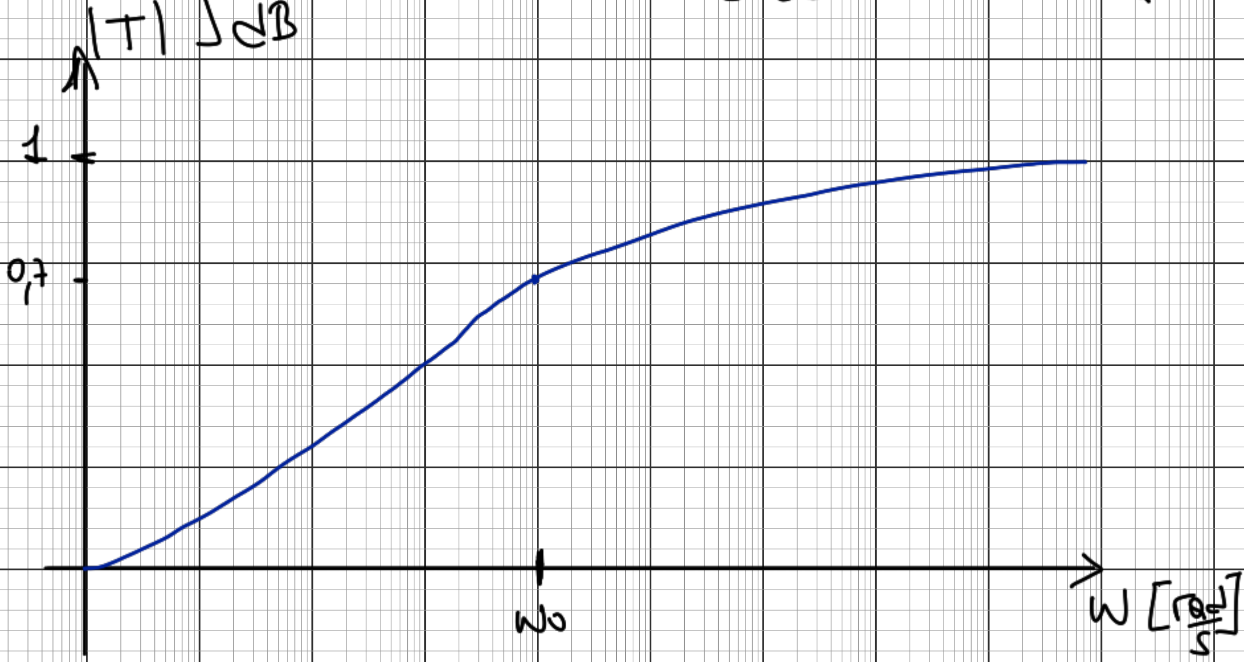



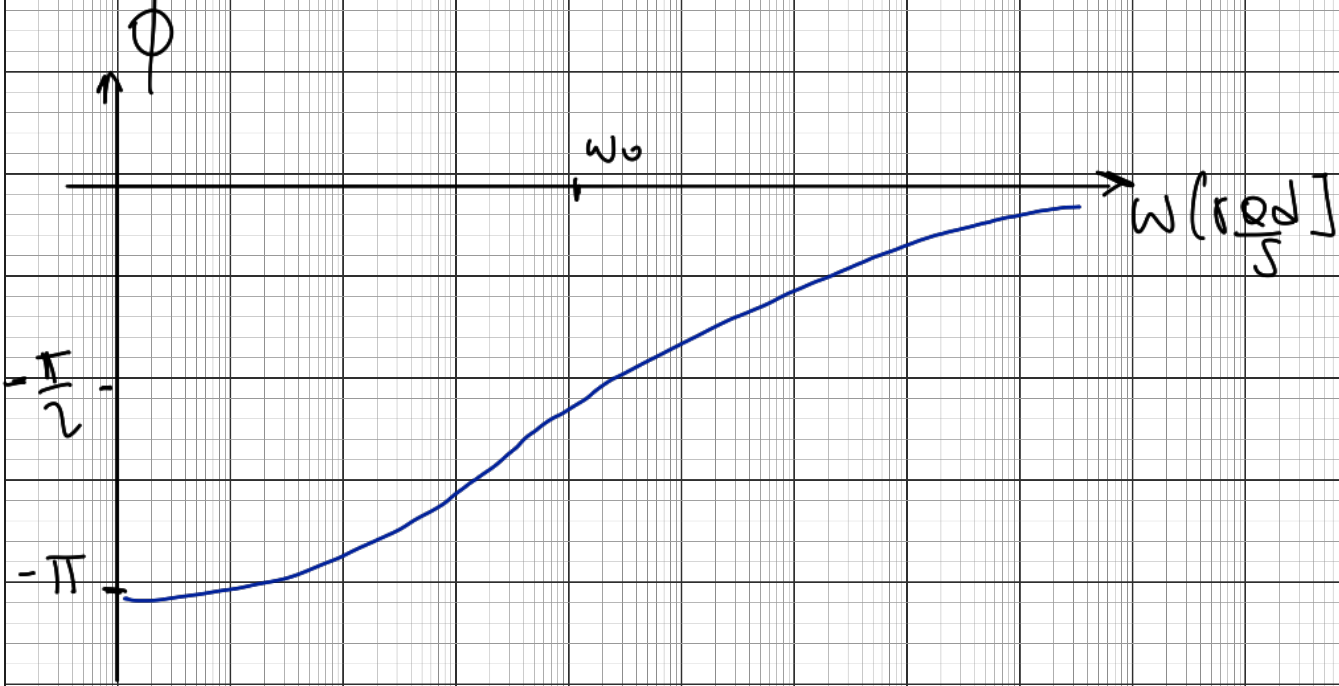
$$T_3^{21} = \frac{s^2 LC}{s^2 LC + 1} \qquad w_0 = \frac{1}{\sqrt{LC}}$$



$$T_{32}^{n} = \frac{s^2 LC}{s^2 LC + 1} \qquad \omega_0 = \frac{1}{\sqrt{LC}}$$



averiguo si son complementarias

- $\left|T^{21}_{13} + T^{21}_{31}\right| = 1$

$$\left|\frac{1}{s^2LC+1} + \frac{s^2LC}{s^2LC+1}\right| = \left|\frac{1+s^2LC}{s^2LC+1}\right| = 1$$ verifica

- $\left|T^{21}_{13} + T^{12}_{32}\right| = 1$

$$\left|\frac{1}{s^2LC+1} + \frac{s^2LC}{s^2LC+1}\right| = \left|\frac{1+s^2LC}{s^2LC+1}\right| = 1$$ verifica

- $\left|T^{21}_{31} + T^{12}_{32}\right| = 1$

$$\left|\frac{s^2LC}{s^2LC+1} + \frac{s^2LC}{s^2LC+1}\right| = \left|2\frac{s^2LC}{s^2LC+1}\right| \neq 1$$ No verifica

(3) a)

$$MAI \begin{pmatrix} s(C_{gs}+C_{gd}) & -sC_{gd} & -sC_{gs} \\ g_m - sC_{gd} & s(C_{gd}+C_{ds}) & -g_m - sC_{ds} \\ -sC_{gs} - g_m & -sC_{ds} & s(C_{gs}+C_{ds})+g_m \end{pmatrix}$$

Source comun

$V_S = 0$

$$\Rightarrow MAD = \begin{pmatrix} s(C_{gs}+C_{gd}) & -sC_{gd} \\ g_m - sC_{gd} & s(C_{gd}+C_{ds}) \end{pmatrix}$$

Drain comun

$V_D = 0$

$$MAD = \begin{pmatrix} s(C_{gs}+C_{gd}) & -sC_{gs} \\ -sC_{gs} - g_m & s(C_{gs}+C_{ds})+g_m \end{pmatrix}$$

(b)

$$\begin{cases} I_1 = y_{11}V_1 + y_{12}V_2 \\ I_2 = y_{21}V_1 + y_{22}V_2 \end{cases} \qquad \begin{cases} V_1 = A V_2 + B(-I_2) \\ I_1 = C V_2 + D(-I_2) \end{cases}$$

$$\frac{I_2 - y_{22}V_2}{y_{21}} = V_1$$

$$I_1 = \frac{y_{11}}{y_{21}}\left(I_2 - y_{22}V_2\right) + y_{12}V_2$$

$$I_1 = \frac{y_{11}}{y_{21}} I_2 + \frac{V_2}{y_{21}}\underbrace{\left(y_{12}y_{21} - y_{11}y_{22}\right)}_{-\Delta Y}$$

$$I_1 = V_2\left(-\frac{\Delta Y}{y_{21}}\right) + \frac{y_{11}}{y_{21}} I_2$$

$$V_1 = -\frac{y_{22}}{y_{21}}V_2 + \frac{1}{y_{21}} I_2$$

$$A = -\frac{y_{22}}{y_{21}} \qquad B = -\frac{1}{y_{21}}$$

$$C = -\frac{\Delta Y}{y_{21}} \qquad D = -\frac{y_{12}}{y_{21}}$$

$$Y \longrightarrow T = \begin{pmatrix} -\frac{y_{22}}{y_{21}} & -\frac{1}{y_{21}} \\ -\frac{\Delta Y}{y_{21}} & -\frac{y_{11}}{y_{21}} \end{pmatrix}$$

$$\begin{pmatrix} s(C_{gs} + C_{gd}) & -sC_{gd} \\ g_m - sC_{gd} & s(C_{gd} + C_{ds}) \end{pmatrix}$$

$$y_{21} = g_m - sC_{gd} \qquad y_{11} = s(C_{gs} + C_{gd}) \qquad y_{22} = s(C_{gd} + C_{ds})$$

$$\Delta Y = s^2(C_{gs} + C_{gd})(C_{gd} + C_{ds}) + sC_{gd}(g_m - sC_{gd})$$

$$\boxed{T_{ABCD} \begin{pmatrix} \dfrac{-s(C_{gd} + C_{ds})}{g_m - sC_{gd}} & -\dfrac{1}{g_m - sC_{gd}} \\ \dfrac{-\left(s^2(C_{gs} + C_{gd})(C_{gd} + C_{ds}) + sC_{gd}(g_m - sC_{gd})\right)}{g_m - sC_{gd}} & -\dfrac{s(C_{gs} + C_{gd})}{g_m - sC_{gd}} \end{pmatrix}}$$